

Appendix A

Appendix

A.1 Glossary of Stochastic Calculus

For a detailed exposition of basic concepts and results on probability theory and stochastic processes we refer to standard text books, for example, Kallenberg (2002), Karatzas & Shreve (1991), Ikeda & Watanabe (1981), Revuz & Yor (1998). In this appendix we give a concise collection of necessary stochastic tools used in this book.

A.1.1 Stochastic Processes in Continuous Time

In this appendix, $(\Omega, \mathcal{F}, P)$ is a probability space with filtration $\mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T_\infty \leq \infty}$ satisfying the “usual conditions”, i.e., continuous from the right, $\bigcap_{s>t} \mathcal{F}_s = \mathcal{F}_t$, and $\mathcal{F}_0$ contains all sets $N \in \mathcal{F}$ with $P(N) = 0$.

A (continuous time) stochastic process X in $\mathbb{R}^m$ is a map $X : \mathbb{R}_+ \times \Omega \ni (t, \omega) \rightarrow \mathbb{R}^m$, such that for each $t \geq 0$ the random variable $X(t) : \omega \rightarrow X(t, \omega)$ is $\mathcal{F}_t/\mathcal{B}(\mathbb{R}^m)$ measurable, where $\mathcal{B}(\mathbb{R}^m)$ denotes the Borel σ -algebra generated by the open subsets of $\mathbb{R}^m$. If, in addition, the process X is left continuous, it may be called *predictable*.

The mapping

$$(k; t_1, \dots, t_k; A_1, \dots, A_k) \longrightarrow P(\{X(t_1) \in A_1, \dots, X(t_k) \in A_k\}), \quad (\text{A.1})$$

for $k \in \mathbb{N}$, $0 \leq t_1 < \dots < t_k$, and $A_1, \dots, A_k$ in $\mathcal{B}(\mathbb{R}^m)$, is called the finite dimensional distribution of the process X . As a general result, the law or distribution of X , being the image probability measure of P on the space of sample paths, is completely determined by the finite dimensional distribution (A.1). Furthermore, the distribution of X is already determined by taking the Borel sets A_i , $i = 1, \dots, k$, in (A.1) of the form $A_i = \{x \in \mathbb{R}^m : x_p \leq a_p^{(i)}, \quad p = 1, \dots, m\}$, $a^{(i)} \in \mathbb{R}^m$.

A.1.1.1 Wiener process or Brownian motion

Due to Wiener’s theorem there exists an $\mathbb{R}$ -valued process W on a probability space $(\Omega, \mathcal{F}, P)$ with $W(0) = 0$, such that for each $\omega \in \Omega$ the sample path $\mathbb{R}_+ \ni t \rightarrow W(t, \omega) \in \mathbb{R}$ is continuous with probability one, and such that

the law of W has finite dimensional distribution given by

$$P\left(\{W(t_1) \leq a^{(1)}, \dots, W(t_k) \leq a^{(k)}\}\right) = \int_{-\infty}^{a^{(1)}} \frac{dx_1}{\sqrt{2\pi t_1}} e^{-\frac{x_1^2}{2t_1}} \cdot \int_{-\infty}^{a^{(2)}} \frac{dx_2}{\sqrt{2\pi(t_2 - t_1)}} e^{-\frac{(x_2 - x_1)^2}{2(t_2 - t_1)}} \dots \int_{-\infty}^{a^{(k)}} \frac{dx_k}{\sqrt{2\pi(t_k - t_{k-1})}} e^{-\frac{(x_k - x_{k-1})^2}{2(t_k - t_{k-1})}}, \quad (\text{A.2})$$

for $k \in \mathbb{N}$, $0 < t_1 < \dots < t_k$, $a^{(i)} \in \mathbb{R}$, $i = 1, \dots, k$.

As a consequence, the process W in (A.2), called *standard Brownian motion* or *Wiener process*, is an almost sure continuous Gaussian process with $W(0) = 0$ and $W(t)$ distributed according to

$$P(\{W(t) \leq a\}) = \int_{-\infty}^a \frac{dx}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}} =: \mathcal{N}(0, t)(a), \quad a \in \mathbb{R},$$

for $t > 0$, which has *independent increments*, i.e., $W(t)$ and $W(t+h) - W(t)$ are independent for $t, h > 0$. Moreover W is completely characterised by these properties. A standard Brownian motion in $\mathbb{R}^m$ is a process $W = (W_1, \dots, W_m)$, where $W_1, \dots, W_m$ are independent one-dimensional Brownian motions.

A.1.2 Martingales and Stopping Times

A *stopping time* τ with respect to a filtration $(\Omega, \mathcal{F})$ is an $\mathbb{R}_+$ -valued random variable for which $\{\tau \leq t\} \in \mathcal{F}_t$ for every $t \geq 0$. Obviously, a stopping time τ is measurable with respect to the associated σ -algebra

$$\mathcal{F}_\tau := \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t, \quad t \geq 0\}.$$

A process M on the probability space $(\Omega, \mathcal{F}, P)$ is called a *martingale*, *submartingale*, or *supermartingale*, if for each $t \geq 0$, $E|M(t)| < \infty$ and, respectively,

$$\begin{aligned} E^{\mathcal{F}_s} M(t) &= M(s), \\ E^{\mathcal{F}_s} M(t) &\geq M(s), \\ E^{\mathcal{F}_s} M(t) &\leq M(s), \quad 0 \leq s \leq t \leq T_\infty. \end{aligned} \quad (\text{A.3})$$

The process M is said to be a *local martingale* if there exists a sequence of stopping times $(\tau_n)_{n \in \mathbb{N}}$ with $\tau_n \uparrow T_\infty$, such that for each n the stopped process $t \rightarrow M(t \wedge \tau_n)$ is a martingale in the sense of (A.3).

A *semi-martingale* X on $(\Omega, \mathcal{F}, P)$ is a right continuous process which can be written as

$$X = X(0) + M + A, \quad (\text{A.4})$$

where M is a local martingale with $M(0) = 0$, and A is a process of locally finite variation starting at zero. Hence, $A(0) = 0$, and for each fixed $t > 0$ we have

$$\sup_{\pi_t \in \mathcal{P}_{[0,t]}} \sum_{i=1}^N |A(t_i) - A(t_{i-1})| < \infty \quad a.s.,$$

where $\mathcal{P}_{[0,t]}$ denotes the set of partitions π_t on $[0, t]$, defined by the sequence $0 = t_0 < t_1 < \cdots < t_N = t$. For any semimartingale the decomposition (A.4), called *Doob–Meyer decomposition*, is unique.

REMARK A.1 The analogue definition of martingale, submartingale, and supermartingale for a discrete parameter processes $(M_n)_{n \in \mathbb{N}_0}$ is clear from (A.3). Moreover, if $(X_n)_{n \in \mathbb{N}}$ is a discrete parameter process adapted to a discrete filtration $(\mathcal{F}_n)_{n \in \mathbb{N}_0}$, it is easy to see that there exists a unique martingale M with $M_0 = 0$, and a process A with $A_0 = 0$ and A_n being $\mathcal{F}_{n-1}$ measurable for $n \geq 1$, such that $X = X_0 + M + A$. In particular, we have the discrete Doob–Meyer decomposition

$$X_n = X_0 + \underbrace{\sum_{k=0}^{n-1} (X_{k+1} - E^{\mathcal{F}_k} X_{k+1})}_M + \underbrace{\sum_{k=0}^{n-1} (E^{\mathcal{F}_k} X_{k+1} - X_k)}_A, \quad n \geq 1.$$

Due to the measurability of A_n with respect to $\mathcal{F}_{n-1}$, A is called predictable. For continuous time processes, however, the notion of predictability is much more complicated. $\square$

A.1.3 Quadratic Variation and the Itô Stochastic Integral

Throughout this book we consider as integrators for stochastic integrals only semimartingales which are continuous, i.e., semimartingales which have almost sure continuous sample paths. We first introduce the notion of covariation or bracket process and quadratic variation process for continuous semimartingales and then recall the Itô stochastic integral and his main properties.

A.1.3.1 The covariation process

Let M_1 and M_2 be continuous local martingales with $M_1(0) = M_2(0) = 0$. Then the *covariation process* or *bracket process* $\langle M_1, M_2 \rangle$ of M_1 and M_2 is defined as the finite variation part of the Doob–Meyer decomposition (A.4) of $M_1 M_2$; hence, $\langle M_1, M_2 \rangle(0) = 0$ and $M_1 M_2 - \langle M_1, M_2 \rangle$ is a local martingale. If X_1 and X_2 are semimartingales with Doob–Meyer decompositions $X_1 = X_1(0) + M_1 + A_1$ and $X_2 = X_2(0) + M_2 + A_2$ according to (A.4), then the covariation process $\langle X_1, X_2 \rangle$ of X_1 and X_2 is defined to be $\langle X_1, X_2 \rangle := \langle M_1, M_2 \rangle$, hence the covariation of the martingale parts of X_1

and X_2 . For a continuous semi-martingale X , the *quadratic variation* $\langle X \rangle$ of X is defined to be $\langle X \rangle := \langle X, X \rangle$. Note that if $X = X(0) + M + A$ for a true martingale M , we have by Jensen's inequality, $E^{\mathcal{F}_s} M^2(t) \geq (E^{\mathcal{F}_s} M(t))^2 = M^2(s)$, hence M^2 is a submartingale. From this it can be shown in general that the quadratic variation process $\langle M, M \rangle = \langle X, X \rangle$ is nondecreasing, and so in particular nonnegative.

It is easy to see that the covariation operator is symmetric and linear with respect to both arguments. Since moreover $\langle X \rangle = \langle X, X \rangle \geq 0$, we have the Cauchy–Schwartz inequality

$$|\langle X_1, X_2 \rangle| \leq \sqrt{\langle X_1, X_1 \rangle} \sqrt{\langle X_2, X_2 \rangle}. \quad (\text{A.5})$$

REMARK A.2 Since the martingale part of a finite variation process is zero, we have the following frequently used fact: If X and A are continuous semimartingales, and A has finite variation, then $\langle X, A \rangle = 0$. $\square$

The terms covariation and quadratic variation are motivated by the following result. Let for $t > 0$, π_t be a partition on $[0, t]$, defined by the sequence, $0 = t_0 < t_1 < \dots < t_N = t$, and let $\delta(\pi_t) := \max_{1 \leq i \leq N} \{t_i - t_{i-1}\}$. Then we have for two continuous semimartingales X_1 and X_2 ,

$$\langle X_1, X_2 \rangle(t) = \lim_{\delta(\pi_t) \rightarrow 0} \sum_{i=1}^N (X_1(t_i) - X_1(t_{i-1}))(X_2(t_i) - X_2(t_{i-1})),$$

where the limit is to be considered in probability.

REMARK A.3 The covariation process $\langle X_1, X_2 \rangle$, which is of finite variation, determines path-wise a measure on $\mathbb{R}_+$. In differential form this measure is usually denoted as $d\langle X_1, X_2 \rangle$, but also as $dX_1 \cdot dX_2$. $\square$

A.1.3.2 The Itô stochastic integral

THEOREM A.1

For a continuous local martingale M and a left continuous (predictable) process ϕ which satisfies $\int_0^t \phi^2(s) d\langle M \rangle(s) < \infty$ a.s. for all $t > 0$, there exists a unique **Itô stochastic integral** process I ,

$$I_{\phi, M}(t) := I(t) := \int_0^t \phi(s) dM(s), \quad (\text{A.6})$$

which satisfies the following properties,

- (i) I is a local martingale with $I(0) = 0$.

(ii) For any continuous semimartingale X we have

$$\langle I, X \rangle = \int_0^t \phi(s) d\langle M, X \rangle(s),$$

where the latter integral is path-wise an ordinary Lebesgue–Stieltjes integral.

By taking $X = I$ in (ii), we have in particular

$$\langle I \rangle = \int_0^t \phi^2(s) d\langle M \rangle(s). \quad (\text{A.7})$$

The Itô integral (A.6) can be generalized to any continuous semimartingale Y as integrator process in the following obvious way. If $Y = Y(0) + M + A$ is the Doob–Meyer decomposition of Y , define

$$I_{\phi, Y}(t) := \int_0^t \phi(s) dY(s) := \int_0^t \phi(s) dM(s) + \int_0^t \phi(s) dA(s), \quad (\text{A.8})$$

provided that the last integral, which is path-wise an ordinary Lebesgue–Stieltjes integral since A is of finite variation, exists almost surely for all $t > 0$. It is easy to see by Remark A.2 that property (ii) in Theorem A.1 and thus (A.7) hold true for (A.8) as well. However, in general the local martingale property (i) is lost of course.

The interpretation of Theorem A.1 is as follows. For a partition $0 = t_0 < t_1 < \dots < t_N < T_\infty \leq \infty$ and a sequence of random variables $(a_j)_{j \in \mathbb{N}_0}$, where a_j is $\mathcal{F}_j$ measurable for $j \geq 0$, consider the elementary predictable step process,

$$\phi(t, \omega) = \sum_{i=1}^N a_{i-1}(\omega) 1_{(t_{i-1}, t_i]}(t). \quad (\text{A.9})$$

By straightforward technique it then follows that the process

$$I(t) := \sum_{i=1}^N a_{i-1} (M(t_i \wedge t) - M(t_{i-1} \wedge t))$$

satisfies (i) and (ii) in Theorem A.1 and (A.7). For general integrands ϕ , the Itô integral (A.6) can be considered as a limit in probability of integrals due to a sequence of elementary step processes (A.9) converging to ϕ .

A.1.3.3 The Itô formula

For the Itô integral process we have the following substitution rule, called *Itô's lemma*.

LEMMA A.1

Let X be a continuous semimartingale in $\mathbb{R}^m$, i.e., $X = (X_1, \dots, X_m)$ where X_i is a continuous semimartingale in $\mathbb{R}$ for $i = 1, \dots, m$, and let $f \in C^2(\mathbb{R}^m)$, i.e., f has continuous partial derivatives of second order. Then, it holds

$$f(X(t)) = f(X(0)) + \sum_{i=1}^m \int_0^t \frac{\partial f}{\partial x_i}(X(s)) dX(s) + \frac{1}{2} \sum_{i,j=1}^m \int_0^t \frac{\partial^2 f}{\partial x_i \partial x_j}(X(s)) d\langle X_i, X_j \rangle(s). \quad (\text{A.10})$$

By taking $f(x, y) = xy$ in Lemma A.1 we obtain the Itô integration by parts formula, also called the Itô product rule,

$$X_1(t)X_2(t) = X_1(0)X_2(0) + \int_0^t X_1(s)dX_2(s) + \int_0^t X_2(s)dX_1(s) + \langle X_1, X_2 \rangle(t). \quad (\text{A.11})$$

A.1.3.4 Itô processes

The continuous stochastic processes involved in Chapters 1-6 are exclusively of Itô type. An Itô type process X in $\mathbb{R}^m$ is defined on a probability space $(\Omega, \mathcal{F}, P)$, where the filtration $\mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T_\infty < \infty}$ is generated by a standard Brownian motion in $\mathbb{R}^m$ which satisfies the “usual conditions”, and can be represented as a stochastic integral,

$$X_i(t) = X_i(0) + \sum_{i=1}^n \int_0^t \alpha_i(s) ds + \sum_{i=1}^n \sum_{j=1}^m \int_0^t \beta_{ij}(s) dW_j(s). \quad (\text{A.12})$$

In (A.12) the processes α_i and β_{ij} are $\mathbb{F}$ predictable (continuous from the left is enough) and for all $t > 0$, $\int_0^t (|\alpha_i(s)| + |\beta_{ij}(s)|) ds < \infty$ a.s. for $1 \leq i \leq n$, $1 \leq j \leq m$.

Of particular importance are positive valued Itô processes of the form

$$Y_i(t) = Y_i(0) \exp \left(-\frac{1}{2} \sum_{i=1}^n \int_0^t |\beta_i|^2(s) ds + \sum_{i=1}^n \int_0^t \alpha_i(s) ds + \sum_{i=1}^n \sum_{j=1}^m \int_0^t \beta_{ij}(s) dW_j(s) \right), \quad (\text{A.13})$$

where $Y_i(0) > 0$, $\beta_i := (\beta_{i1}, \dots, \beta_{im})^\top$ and $|\beta_i|^2 = \beta_i^\top \beta_i$, for $1 \leq i \leq n$. By applying Itô's Lemma A.1 to (A.13) we obtain in differential form

$$\frac{dY_i(t)}{Y_i(t)} = \sum_{i=1}^n \alpha_i(t) dt + \sum_{i=1}^n \sum_{j=1}^m \beta_{ij}(t) dW_j(t). \quad (\text{A.14})$$

The component solutions (A.13) of (A.14) are usually called Doléans exponentials and denoted as $Y_i = \mathcal{E}(X_i)$, where X_i is given by (A.12) for $1 \leq i \leq n$. The next lemma yields a convenient formula for the ratio of two Doléans exponentials.

LEMMA A.2

Let X and Y be given by

$$\begin{aligned}\frac{dX}{X} &= \mu_X dt + \sigma_X^\top dW, \\ \frac{dY}{Y} &= \mu_Y dt + \sigma_Y^\top dW\end{aligned}$$

respectively. Then we have for X/Y ,

$$\frac{d(X/Y)}{X/Y} = (\mu_X - \mu_Y - \sigma_Y^\top(\sigma_X - \sigma_Y)) dt + (\sigma_X - \sigma_Y)^\top dW. \quad (\text{A.15})$$

In terms of Doléans exponentials, Lemma A.2 reads

$$X = \mathcal{E}(P) \quad \wedge \quad Y = \mathcal{E}(Q) \quad \Rightarrow \quad \frac{X}{Y} = \mathcal{E}(P - Q - \langle P - Q, Q \rangle).$$

A.1.4 Regular Conditional Probability

The concept of regular conditional probability is in fact an extension of the elementary notion of conditional probability, $P(A|B) := P(A \cap B)/P(B)$ for a set B of positive measure, to conditioning sets of measure zero.

Let $\mathcal{G}$ be a sub σ -algebra of $\mathcal{F}$. Then a map

$$\Omega \times \mathcal{F} \ni (\omega, A) \rightarrow P(\omega, A)$$

is called a *regular conditional probability given $\mathcal{G}$* , if

- (i) For fixed $\omega \in \Omega$, $P(\omega, \cdot)$ is a probability measure on $(\Omega, \mathcal{F})$;
- (ii) For fixed $A \in \mathcal{F}$, the random variable $\omega \rightarrow P(\omega, A)$ is $\mathcal{G}$ measurable;
- (iii) For any $\mathcal{F}$ -measurable random variable Z it holds

$$[E^{\mathcal{G}} Z](\omega) = \int P(\omega, d\tilde{\omega}) Z(\tilde{\omega}) \quad a.s.$$

According to a fundamental theorem (see, e.g., Ikeda and Watanabe (1981)) a regular conditional probability given $\mathcal{G} \subset \mathcal{F}$ exists and is unique, if the basic probability space $(\Omega, \mathcal{F}, P)$ is a *standard* probability space. For the definition of a standard probability space, see also Ikeda and Watanabe (1981). Without giving further details we just notice that all probability spaces considered in this book are standard.

A.2 Minimum Search Procedures

The objective functions encountered in the calibration procedures presented in Chapter 3 are yet smooth but may have more local minima. Therefore it is important to have a proper search routine for locating the global minimum of such functions. Although standard search routines such as Powel and Levenberg-Marguardt are basically local methods, they can be advantageously combined with global search methods based on minimization over random points or quasi-random points in the search space. The idea is as follows: Generate a number of starting points $s_1, \dots, s_K$ in the search space and then start from each point a local minimum search, thus yielding a set of local minima $s_1^*, \dots, s_K^*$. Then take as search result $s_{j^*}^*$, where $f(s_{j^*}^*) = \min_{1 \leq j \leq K} f(s_j^*)$ and f is the objective function. For the calibration experiments in Chapter 3, the Powel method is used as local search algorithm (for details see for instance Numerical Recipes in C), whereas quasi-random Halton points (see Section A.2.1) are used for the global searcher.

The advantage of quasi-random numbers over uniformly distributed Monte Carlo numbers is that their distribution in space is more uniform since the generation of these points in space proceeds self-avoiding in a sense. Due to this feature quasi-random points are also particularly suitable for high dimensional integration: If g is some smooth function on $[0, 1]^d$ and $q_1, \dots, q_N$ is a system of quasi-random numbers in $[0, 1]^d$ (for instance Halton or Sobel points), then the accuracy of the estimation

$$\int_{[0,1]^d} g(x)dx \approx \frac{1}{N} \sum_{i=1}^N g(q_i)$$

is asymptotically of order of $O(\log^d N/N)$ which is essentially better than the order $O(1/\sqrt{N})$ error due to (pseudo) Monte Carlo random numbers (for large d the enumerator $\log^d N$ can become quite large however).

A.2.1 Halton Quasi-random Numbers

Halton points can be generated very easily. In one dimension the recipe for the generation of the j -th Halton point, $j \geq 1$, is as follows.

- (i) Choose a prime p .
- (ii) Write j as $j = a_k p^k + a_{k-1} p^{k-1} + \dots + a_1 p + a_0$, for integers $0 \leq a_i < p$, $0 = 1, \dots, k$, with $a_k \neq 0$.
- (iii) Set $H_j^{(p)} = (a_0 p^k + a_1 p^{k-1} + \dots + a_{k-1} p + a_k) p^{-(k+1)}$.

For arbitrary dimension d , $d \geq 1$, we take the first d prime numbers $p_1, \dots, p_d$, and then the j -th d -dimensional Halton vector, $j \geq 1$, is given as

$$H_j = \left(H_j^{(p_1)}, \dots, H_j^{(p_d)} \right).$$

Below we give C++ code for the generation of Halton points.¹ The main program in this code generates, as an example, the first ten Halton points in the five dimensional cube $[0, 1]^5$.

C++ code for generation of Halton points

```
class CHalton{
public:
    CHalton();
    virtual ~CHalton();
    bool SetDimension(int dimension); //returns true, if no error occurred
    void GetHaltonVector(int i, double *x);
        // x points to the Storage, where dimension
        // double values between 0 and 1 will be stored.
        // i should be >=1.
    double Halton(int i, int base) const;
protected:
    double FastHalton(int i, int base) const;
private:
    int m_DIM;
    int *m_prime;
};

CHalton::CHalton(){
    m_DIM=0;
    m_prime=NULL;
}

CHalton::~~CHalton(){
    if (m_prime!=NULL) delete [] m_prime;
}

bool CHalton::SetDimension(int dimension){
    if (m_prime!=NULL) delete [] m_prime;
    if (dimension <= 0){
        m_DIM=0;
        m_prime=0;
        return dimension == 0;
    }
}
```

¹This code is written by O. Reiß.

```

    }
    m_prime=new int [dimension];
    if (m_prime==NULL){
        m_DIM=0;
        m_prime=0;
        return false;
    }
    m_DIM=dimension;
    int i,z=1;
    bool is_prim;
    for(dimension=0;dimension<m_DIM;dimension++){
        //Search the next prime
        do{
            z++;
            i=0;
            is_prim=true;
            while (is_prim && i<dimension){
                is_prim=is_prim && (z%m_prime[i]);
                i++;
            }
        }while (!is_prim);
        m_prime[dimension]=z;
    }
    return true;
}

void CHalton::GetHaltonVector(int i,double *x){
    int d;
    if (i<=0){
        for(d=0;d<m_DIM;d++) x[d]=0.0;
        return;
    }
    for(d=0;d<m_DIM;d++){
        x[d]=FastHalton(i,m_prime[d]);
    }
}

double CHalton::Halton(int i,int base)const{
    if (i<=0 || base <=1) return 0.0;
    return FastHalton(i,base);
}

double CHalton::FastHalton(int i,int base)const{
    const double y=1.0/base;
    double x=y;

```

```

double erg=0.0;
while(i!=0){
    erg+=x*(double)(i%base);
    i/=base;
    x*=y;
}
return erg;
}

void main(){
    CHalton hal;
    int DIM=5;
    int NumberOfHaltonPoints=10;
    int i,d;
    double *x;
    hal.SetDimension(DIM);
    x=new double[DIM];
    for(i=1;i<=NumberOfHaltonPoints;i++){
        hal.GetHaltonVector(i,x);
        for(d=0;d<DIM;d++){
            cout << x[d] << "\t";
        }
        cout << endl;
    }
    delete [ ] x;
}

```

A.3 Additional Proofs

A.3.1 Covariance of Two Black-Scholes Models

Consider two correlated Black-Scholes models driven by two independent Brownian motions W_1 and W_2 ,

$$\begin{aligned}\frac{dS_1}{S_1} &= \mu_1 dt + \sigma_1 dW_1 \\ \frac{dS_2}{S_2} &= \mu_2 dt + \sigma_2 (\rho dW_1 + \sqrt{1-\rho^2} dW_2),\end{aligned}$$

where $-1 \leq \rho \leq 1$. Then by multiplying the corresponding Doléans exponentials,

$$\begin{aligned} \frac{ES_1(T)S_2(T)}{S_1(0)S_2(0)} &= E \exp \left[\left(\mu_1 + \mu_2 - \frac{\sigma_1^2}{2} - \frac{\sigma_2^2}{2} \right) T \right. \\ &\quad \left. + (\sigma_1 + \sigma_2 \rho) W_1(T) + \sigma_2 \sqrt{1 - \rho^2} W_2(T) \right] \\ &= E \exp \left[\left(\mu_1 + \mu_2 - \frac{\sigma_1^2}{2} - \frac{\sigma_2^2}{2} \right) T + \sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2} W_1(T) \right] \\ &= \exp [\mu_1 T + \mu_2 T + \rho\sigma_1\sigma_2 T], \end{aligned}$$

where we use that $\exp[-a^2 T/2 + aW(T)]$ is a martingale. It then easily follows that

$$\begin{aligned} Cov(S_1(T), S_2(T)) &= ES_1(T)S_2(T) - (ES_1(T))(ES_2(T)) \\ &= S_1(0)S_2(0) \exp [\mu_1 T + \mu_2 T + \rho\sigma_1\sigma_2 T] \\ &\quad - S_1(0)S_2(0) \exp [\mu_1 T + \mu_2 T] \\ &\approx S_1(0)S_2(0)\rho\sigma_1\sigma_2 T, \end{aligned}$$

when T is not too large.

A.3.2 Proof of Equality (5.26)

We here prove the equality (5.26). Let us write

$$Y^{m+1(i)} - Y^m(i) = 1_{\tau_i^{(m+1)} = i} \left(Y^{m+1(i)} - Y^m(i) \right) + 1_{\tau_i^{(m+1)} > i} \left(Y^{m+1(i)} - Y^m(i) \right). \quad (\text{A.16})$$

Note that the first term in (A.16) is zero. Indeed, if $\tau_i^{(m+1)} = i$, then $Y^{m+1(i)} = E^{\mathcal{F}^{(i)}} Z^{(\tau_i^{(m+1)})} = Z^{(i)}$, and on the other hand, $Z^{(i)} \leq Y^m(i) \leq Y^{m+1(i)}$. Now we consider the second term in (A.16). If $\tau_i^{(m+1)} > i$, then $\tau_i^{(m+1)} = \tau_{i+1}^{(m+1)}$ and so $\tau_i^{(m+1)} > i \Rightarrow Y^{m+1(i)} = E^{\mathcal{F}^{(i)}} Z^{(\tau_i^{(m+1)})} = E^{\mathcal{F}^{(i)}} Z^{(\tau_{i+1}^{(m+1)})} = E^{\mathcal{F}^{(i)}} Y^{m+1(i+1)}$.

We thus obtain,

$$\begin{aligned}
 & Y^{m+1}(i) - Y^m(i) \\
 &= 1_{\tau_i^{(m+1)} > i} \left(E^{\mathcal{F}^{(i)}} Y^{m+1}(i+1) - Y^m(i) \right) \\
 &= E^{\mathcal{F}^{(i)}} 1_{\tau_i^{(m+1)} > i} (Y^{m+1}(i+1) - Y^m(i+1)) + E^{\mathcal{F}^{(i)}} 1_{\tau_i^{(m+1)} > i} (Y^m(i+1) - Y^m(i)) \\
 &= E^{\mathcal{F}^{(i)}} 1_{\tau_i^{(m+1)} > i} \left(E^{\mathcal{F}^{(i+1)}} 1_{\tau_{i+1}^{(m+1)} > i+1} (Y^{m+1}(i+2) - Y^m(i+2)) \right. \\
 &\quad \left. + E^{\mathcal{F}^{(i+1)}} 1_{\tau_{i+1}^{(m+1)} > i+1} (Y^m(i+2) - Y^m(i+1)) \right) \\
 &\quad + E^{\mathcal{F}^{(i)}} 1_{\tau_i^{(m+1)} > i} (Y^m(i+1) - Y^m(i)) \\
 &= E^{\mathcal{F}^{(i)}} 1_{\tau_i^{(m+1)} > i+1} (Y^{m+1}(i+2) - Y^m(i+2)) + \\
 &\quad + E^{\mathcal{F}^{(i)}} 1_{\tau_i^{(m+1)} > i+1} (Y^m(i+2) - Y^m(i+1)) \\
 &\quad + E^{\mathcal{F}^{(i)}} 1_{\tau_i^{(m+1)} > i} (Y^m(i+1) - Y^m(i)).
 \end{aligned}$$

Now, by induction from i to k and the fact that $Y^{m+1}(k) = Y^m(k)$, it follows that

$$\begin{aligned}
 Y^{m+1}(i) - Y^m(i) &= E^{\mathcal{F}^{(i)}} \sum_{p=i}^{k-1} 1_{\tau_i^{(m+1)} > p} \left(Y^m(p+1) - Y^m(p) \right) \\
 &= E^{\mathcal{F}^{(i)}} \sum_{p=i}^{\tau_i^{(m+1)} - 1} 1_{\tau_p^{(m)} = p} \left(E^{\mathcal{F}^{(p)}} Y^m(p+1) - Y^m(p) \right),
 \end{aligned}$$

since $E^{\mathcal{F}^{(p)}} Y^m(p+1) = Y^m(p)$ on the set $\{\tau_p^{(m)} > p\}$.

A.3.3 Proof of Proposition 5.2

For all i , $0 \leq i \leq k$, and $m = 0, 1, 2, \dots$, satisfying $m \geq k - i$ we have almost surely

$$Y^{m+1}(i) = Y^m(i),$$

i.e., the sequence becomes constant. We will prove this by backward induction over i . For $i = k$ the statement is true since for all $m \geq 0$,

$$Y^m(k) = Z^{(k)}.$$

Suppose now the statement is already proved for all j , $i + 1 \leq j \leq k$. By Theorem 5.1 we have for $0 \leq j \leq k$ and $m \geq 1$,

$$\begin{aligned}
 Y^m(j) &\geq \tilde{Y}^m(j) = \max_{p: j \leq p \leq k} E^{\mathcal{F}^{(j)}} Z^{(\tau_p^{m-1})} \\
 &\geq E^{\mathcal{F}^{(j)}} Z^{(\tau_{j+1}^{m-1})} = E^{\mathcal{F}^{(j)}} E^{\mathcal{F}^{(j+1)}} Z^{(\tau_{j+1}^{m-1})} \\
 &= E^{\mathcal{F}^{(j)}} Y^{m-1}(j+1).
 \end{aligned} \tag{A.17}$$

So, by Proposition 5.26 and the quasi-supermartingale property (A.17) we obtain for $m \geq 1$,

$$\begin{aligned}
 Y^{m+1}(i) - Y^m(i) &= E^{\mathcal{F}^{(i)}} \sum_{p=i}^{\tau_i^{(m+1)}-1} E^{\mathcal{F}^{(p)}} Y^m(p+1) - Y^m(p) \\
 &\leq E^{\mathcal{F}^{(i)}} \sum_{p=i}^{\tau_i^{(m+1)}-1} E^{\mathcal{F}^{(p)}} Y^m(p+1) - E^{\mathcal{F}^{(p)}} Y^{m-1}(p+1) \\
 &\leq E^{\mathcal{F}^{(i)}} \sum_{p=i+1}^k Y^m(p) - Y^{m-1}(p), \tag{A.18}
 \end{aligned}$$

where the terms in the sum are non-negative because $Y^m \geq Y^{m-1}$ due to Theorem 5.1 again. Suppose now that $m \geq k - i$. Hence, for all $p \geq i + 1$, we have $m - 1 \geq k - p$. Then, by the induction hypothesis, we have $m \geq 1$ and $Y^{m-1}(p) = Y^m(p)$ for all $p \geq i + 1$. Hence from (A.18) it follows that $Y^{m+1}(i) - Y^m(i) = 0$.

A.3.4 Proof of Proposition 5.3

$$\begin{aligned}
 \frac{V_0^{up,K}}{B(0)} &= E \sup_{1 \leq j \leq k} \left[\frac{C_{\mathcal{T}_j}}{B(\mathcal{T}_j)} - \sum_{q=1}^j \tilde{Y}^{(q)} + \sum_{q=1}^j \frac{1}{K} \sum_{i=1}^K \xi_i^{(q)} \right] \\
 &= E \left[\frac{C_{\mathcal{T}_{\hat{j}_{\max}}}}{B(\mathcal{T}_{\hat{j}_{\max}})} - \sum_{q=1}^{\hat{j}_{\max}} \tilde{Y}^{(q)} + \sum_{q=1}^{\hat{j}_{\max}} \frac{1}{K} \sum_{i=1}^K \xi_i^{(q)} \right] \\
 &= E (1 - 1_{[j_{\max} \neq \hat{j}_{\max}]}) \left[\frac{C_{\mathcal{T}_{j_{\max}}}}{B(\mathcal{T}_{j_{\max}})} - \sum_{q=1}^{j_{\max}} \tilde{Y}^{(q)} + \sum_{q=1}^{j_{\max}} \frac{1}{K} \sum_{i=1}^K \xi_i^{(q)} \right] \\
 &\quad + E 1_{[j_{\max} \neq \hat{j}_{\max}]} \left[\frac{C_{\mathcal{T}_{\hat{j}_{\max}}}}{B(\mathcal{T}_{\hat{j}_{\max}})} - \sum_{q=1}^{\hat{j}_{\max}} \tilde{Y}^{(q)} + \sum_{q=1}^{\hat{j}_{\max}} \frac{1}{K} \sum_{i=1}^K \xi_i^{(q)} \right] \\
 &= \frac{V_0^{up,K}}{B(0)} + O((P(j_{\max} \neq \hat{j}_{\max}))^{1-1/q})
 \end{aligned}$$

by Hölder's inequality, for any integer q . The latter assertion is due to the fact that for any q the q -th moment of both

$$\begin{aligned} \frac{C_{\mathcal{T}_{\hat{j}_{\max}}}}{B(\mathcal{T}_{\hat{j}_{\max}})} - \sum_{q=1}^{\hat{j}_{\max}} \tilde{Y}^{(q)} + \sum_{q=1}^{\hat{j}_{\max}} \frac{1}{K} \sum_{i=1}^K \xi_i^{(q)} \\ \text{and} \end{aligned} \quad (\text{A.19})$$

$$\frac{C_{\mathcal{T}_{j_{\max}}}}{B(\mathcal{T}_{j_{\max}})} - \sum_{q=1}^{j_{\max}} \tilde{Y}^{(q)} + \sum_{q=1}^{j_{\max}} \frac{1}{K} \sum_{i=1}^K \xi_i^{(q)}$$

exist and are uniformly bounded in K (we omit the proof). Then, since $\lim_{K \rightarrow \infty} P(j_{\max} \neq \hat{j}_{\max}) = 0$, the convergence for $K \rightarrow \infty$ of $V_0^{up^{up}, K} \rightarrow V_0^{up, K}$ follows.

The convergence $V_0^{up_{low}, K} \rightarrow V_0^{up, K}$ can be shown in a similar way.

